

Lecture 23: Mode splitting and coupling

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- A key result, the **diagonal sum rule**, will show us how to use observations of split modes to learn about the Earth's average radial structure.
- This provides the primary method for constraining the Earth's density profile.

Modes in a Spherical Earth: Two Families

In a spherically symmetric Earth, the normal mode eigenfunctions separate into two distinct families based on their motion.

Toroidal Modes

- Purely tangential motion.
- Like twisting a rubber ball.
- No radial motion, no volume change.
- No perturbation to the gravitational field.
- Cannot exist in fluid regions (like the outer core).

Spheroidal Modes

- Both radial and tangential motion.
- Like breathing or squeezing a ball.
- Involve compression and dilatation.
- Perturb the gravitational field.
- Can exist in both solid and fluid regions.

Mode Classification and Degeneracy

Each normal mode is identified by a set of integer indices:

- n : The **overtone number** ($n = 0, 1, 2, \dots$), related to the number of radial nodes.
- l : The **angular degree** ($l = 0, 1, 2, \dots$), related to the number of nodal lines on the surface.
- m : The **azimuthal order** ($-l \leq m \leq l$), related to the pattern of motion around the rotation axis.

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Degeneracy

In a perfectly spherical, non-rotating Earth, the frequency of a mode depends on n and l , but **not on m** .

- For a given n and l , there are $2l + 1$ modes (one for each m) that all share the exact same frequency.
- This is a $(2l + 1)$ -fold **degeneracy**.
- The set of these $2l + 1$ modes is called a **multiplet** (e.g., ${}_0S_2$). The individual modes within it are called **singlets**.

Breaking the Degeneracy: Perturbation Theory

The Earth is not perfectly spherical or non-rotating. These aspherical perturbations break the degeneracy.

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Result

The frequency shifts for the $2l + 1$ singlets within a multiplet are found by solving a $(2l + 1) \times (2l + 1)$ matrix eigenvalue problem.

The Splitting Matrix

The perturbation theory leads to the following eigenvalue problem for the frequency shifts $\delta\omega$:

The Eigenvalue Problem for Splitting

$$\mathbf{H}'\mathbf{a} = \delta\omega\mathbf{a}$$

- $\mathbf{H}'$ is a $(2l + 1) \times (2l + 1)$ Hermitian matrix whose elements measure the interaction of the perturbation with the modes.

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- The $2l + 1$ eigenvalues of this matrix are the frequency shifts $\delta\omega$ for each of the singlets. The eigenvectors $\mathbf{a}$ describe how the original degenerate eigenfunctions mix.
- In general, the eigenvalues will be distinct, so the multiplet "splits" into $2l + 1$ observable peaks in the spectrum.

Selection Rules

Many of the elements in the splitting matrix are zero due to symmetry. The conditions for an element to be non-zero are called **selection rules**.

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Analogy to Quantum Mechanics

These rules are mathematically identical to the selection rules governing transitions and interactions in quantum systems involving angular momentum.

The Diagonal Sum Rule

A key result from splitting theory is the **diagonal sum rule** (Gilbert, 1971).

The Rule

The trace of the splitting matrix $\mathbf{H}'$ is zero. Since the trace of a matrix is the sum of its eigenvalues, this means:

$$\sum_{m=-l}^l \delta\omega_m = 0$$

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Practical Implication

This gives us a powerful tool. We can observe the frequencies of the split singlets. By taking their arithmetic average, we can recover the frequency of the unsplit multiplet, which is the frequency the Earth would have if it were perfectly spherical.

Constraining the Earth's Density

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A Key Result

By jointly inverting travel-time data and normal mode frequency data, it became possible to independently solve for the Earth's radial velocity and density profiles. This is how we know the density of the core and mantle.

Lateral Density Variations and Mode Coupling

A major goal of modern seismology is to map 3D variations in density, which is crucial for understanding mantle dynamics.

- For example, are the Large Low Shear Velocity Provinces (LLSVPs) at the base of the mantle hot and buoyant, or are they compositionally distinct and dense?

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- Simple perturbation theory is often not accurate enough for this, because it ignores the coupling between different multiplets.
- Modern research uses **mode coupling theory**, which solves a much larger eigenvalue problem that includes the interactions between nearby multiplets.
- This is computationally very expensive, and constraining 3D density variations remains a challenging frontier of seismology.

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- **Selection rules**, based on spherical harmonic properties, determine which perturbations can cause splitting and coupling.
- The **diagonal sum rule** states that the average frequency of a split multiplet is equal to the unsplit frequency of the reference spherical Earth.
- This rule was crucial for using observed mode frequencies to constrain the Earth's radial **density** structure.